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PLL USING A DCO PERIOD LENGTH FEEDBACK CLOCK PULSE TO DETERMINE PHASE ERROR AND TDC GAIN

Abstract

A phase-locked loop (PLL) includes a digitally controlled oscillator (DCO) that generates a DCO output signal. A feedback divider is coupled to the DCO output signal, divides the DCO output signal, and generates a feedback clock signal. The feedback clock signal is generated as a pulse having a pulse width equal to one period of the DCO output signal. A time-to-digital converter (TDC) receives a reference clock signal and the feedback clock signal and generates a TDC output that indicates a phase difference between the reference clock signal and the feedback clock signal. The TDC output also provides gain information indicating how many delay elements correspond to the period of the DCO output signal. The gain information can be used to more accurately cancel quantization error in the PLL.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

[0001] This application relates to application serial number, having attorney docket number 026-0482, naming Sheng Jue Peh as inventor, filed the same day as the present application, and entitled “Feedback Divider in a PLL Used As A Phase/Frequency Detector”, which application is incorporated herein by reference in its entirety.

BACKGROUND

Field of the Invention

[0002] This disclosure relates to phase-locked loops (PLLs) and more particularly to utilization of a time-to-digital converter (TDC) to provide phase error and TDC gain information.

Description of the Related Art

[0003] FIG. **1** illustrates a high level block diagram of a conventional digital PLL **100**. The PLL **100** includes a time-to-digital converter (TDC) **102** that supplies a phase difference (also referred to herein as phase error) **104** between the reference clock (fref) **106** and feedback clock signal (fb_clk) **108**. The phase difference **104** is supplied to the digital loop filter **110** that supplies a control signal **112** to a digitally controlled oscillator (DCO) **114**. The DCO **114** supplies an output signal **116** to the feedback divider **118**, which in turn divides the DCO_OUT signal **116** to generate fb_clk **108**. FIG. **2** illustrates a phase difference at **202** that can occur between fref **106** and fb_clk **108**. Note that both clock signals have the same duty cycle (conventionally 50%) and when aligned have the same rising (and falling) edges. FIG. **3** illustrates a representative output from the TDC when the PLL is locked assuming a 40 bit output. In the illustrated embodiment the TDC output is half 1's and half 0's when locked. That is tdc_out<0:19>=1s and tdc_out <20:39>=0s. The reference clock fref or a delayed version of fref is used to sample the TDC output. The phase information is based on the 0 crossing in the TDC output shown at **302**. Remember that the phase error supplied by the TDC is used to keep the PLL locked. With temperature and/or supply voltage changes, the delay through the delay elements of the TDC may change and number of delay elements that are “1” and “0” when locked may change slightly, e.g., to 19 “1s” and 21 “0s. That change relates to the gain relationship between the TDC output and the DCO output. In conventional PLLs, one way to determine the gain relationship requires taking the PLL out of lock, which means the PLL is no longer supplying a clock signal operationally that is locked to the reference clock signal. Another approach algorithmically checks the output error and adjusts the gain until the error is minimized. Improvements in determining the gain relationship can help improve PLL operation.

SUMMARY OF EMBODIMENTS OF THE INVENTION

[0004] In embodiments, the phase error and gain information from the TDC is determined simultaneously during normal operation of the PLL.

[0005] In an embodiment a method includes recovering phase information from an output of a time-to-digital converter (TDC). The phase information is indicative of a phase difference between a reference clock signal and a feedback clock signal. In addition, gain information is recovered from the output of the TDC. The gain information is indicative of a relationship between a step size of the TDC and a period of a digitally controlled oscillator (DCO) output signal supplied by a digitally controlled oscillator (DCO). In an embodiment, the feedback clock signal has a pulse having a pulse width equal to a period of the DCO output signal.

[0006] In another embodiment a phase-locked loop (PLL) includes a time-to-digital converter (TDC) that is coupled to receive a reference clock signal and a feedback signal. The TDC provides a TDC output that is indicative of a phase difference between the reference clock signal and the feedback signal. A feedback divider is coupled to a digitally controlled oscillator (DCO) output signal and configured to divide the DCO output signal and supply the feedback signal to the TDC.

The TDC output further provides gain information that is indicative of a relationship between a step size of the TDC and a period of the DCO output signal.
[0007] In another embodiment an apparatus includes a digitally controlled oscillator (DCO) that is configured to supply a DCO output signal. A feedback divider is coupled to the DCO output signal and is configured to divide the DCO output signal and supply a feedback signal. The feedback clock signal includes a pulse with a pulse width of a period of the DCO output signal. A time-to-digital converter (TDC) is coupled to receive a reference clock signal and the feedback signal and is configured to provide a TDC output indicative of a phase difference between the reference clock signal and the feedback signal. The phase difference indicated is based on a transition between zeros and ones in the TDC output. The TDC output further provides gain information indicating a relationship between a step size of the TDC and a period of the DCO output clock signal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0008] The present invention may be better understood, and its numerous objects, features, and advantages made apparent to those skilled in the art by referencing the accompanying drawings.
[0009] FIG. **1** illustrates a high level block diagram of a conventional digital PLL.
[0010] FIG. **2** illustrates a timing diagram of the reference clock, the feedback clock, and the TDC output.
[0011] FIG. **3** illustrates the TDC output with the PLL locked.
[0012] FIG. **4** illustrates an embodiment of a PLL in which the TDC simultaneously provides phase error and gain information during normal operation of the PLL.
[0013] FIG. **5** illustrates the feedback divider supplying a feedback clock signal (fb_clk) with a pulse width equal to the period of the DCO_OUT signal.
[0014] FIG. **6** illustrates a block diagram of an embodiment of the TDC.
[0015] FIG. **7** illustrates the feedback clock pulse, the delayed reference clock signal used to sample the TDC and the sampling point for the TDC.
[0016] FIG. **8** illustrates an example of the TDC output.
[0017] FIG. **9** illustrates an embodiment of a first order ΔΣ modulator.
[0018] FIG. **10** illustrates an embodiment of quantization error cancellation using the gain information from the TDC.
[0019] The use of the same reference symbols in different drawings indicates similar or identical items.
DETAILED DESCRIPTION
[0020] Embodiments described herein improve accuracy of PLL operation (for both integer N and fractional-N PLLs) and particularly improve quantization error cancellation in fractional-N PLLs. Embodiments provide continuous calibration of quantization error phase cancellation without extra circuitry or complex algorithms. Both calibration and normal phase detection happens substantially simultaneously as explained further herein, and hence there is no need to disable lock for calibration to happen. Power consumption of the TDC is exactly the same as in conventional TDCs. In addition, the range of the phase detector implemented by the TDC is extended by 1 DCO period since both rising and falling edge information can be used for phase detection, which is significant for an embodiment in which the TDC is a thermometer/unit-weighted design.
[0021] FIG. **4** illustrates an embodiment of a PLL **400** in which the TDC simultaneously provides phase error and gain information during normal operation of the PLL. While the operation of PLL **400** is similar to PLL **100**, differences allow continuous calibration using the gain information from the TDC as well as the phase error between ref_clk and fb_clk being provided to the loop filter to control the DCO and maintain the PLL in lock. The PLL **400** includes a time-to-digital converter

(TDC) **402** that supplies the phase difference **404** between the reference clock (fref) **406** and feedback clock signal (fb_clk) **408**. The phase difference **404** is supplied to the digital loop filter and cancellation logic **410** that supplies a control signal **412** to the DCO **414**. The digital loop filter and cancellation logic **410** is described further herein. The DCO **414** supplies an output signal **416** to the feedback divider **418**, which in turn divides the DCO_OUT signal **416** to generate the feedback clock signal (fb_clk) **408**. However, instead of supplying a feedback clock signal with a 50% duty cycle, the feedback divider **418** supplies a feedback clock signal (fb_clk) as shown in FIG. **5** with a pulse width equal to the period of the DCO_OUT signal. That facilitates the extraction of both phase information and gain information from the TDC output.

[0022] FIG. **6** illustrates a block diagram of an embodiment of the TDC **402**. As known to those of skill in the art, many TDC implementations are known and various embodiments utilize a TDC implementation suitable for the particular embodiment. The TDC **402** shown in FIG. **6** includes a plurality of delay elements **602** and flip-flops **604**. The fb_clk **408** has a pulse width equal to the period of DCO_OUT and is supplied to the delay elements and the flip-flops. A delayed version of the reference clock signal (delayed_fref) **606** samples the TDC and provides the TDC output <Q.sub.0:Q.sub.n-1>. With the rising edge of fb_clk, 1's begin to traverse down the delay elements of the TDC. With the falling edge of the fb_clk, 0's begin to traverse down the delay elements. FIG. **7** illustrates the fb_clk **408** pulse **702**, the rising edge of the delayed fref **606** at **704** resulting the sampling of the TDC at **706** responsive to the rising edge of the delayed fref signal **606**.

[0023] FIG. **8** illustrates an example of the TDC output. FIG. **8** assumes a TDC that supplies 50 sample points. Because the pulse of the feedback clock signal is short (1 DCO period) as shown in FIG. **7**, the first 9 bits tdc_out<0:8>=all 0s reflecting that the falling edge of the feedback pulse occurs before the sample is taken at **706** and 0s start to traverse down the delay elements before the sample is taken. The middle bits of the TDC output tdc<9:40>=all 1s, reflecting the feedback pulse. Finally, the remaining bits in the TDC output tdc<41:49>=all 0s reflecting the condition of the fb_clk before the pulse. The phase information can be extracted at the transitions between 0 and 1 at either or both **802** and **804**. The gain information is indicated by the length of the 1 bit stream at **806**. In the example given, the length of the 1-bit stream is 32 bits. Thus, the 32 1-bit stream corresponds to the period of DCO_OUT. Thus, the period of the DCO_OUT clock corresponds to 32 delay elements of the TDC.

[0024] That gain information is particularly useful in calibrating the cancellation of quantization error in fractional-N PLLs. Fractional-N PLLs allow dividing the DCO_OUT signal by a non-integer number. Referring back to FIG. **4**, the divider control block **420** includes a ΔΣ modulator. The feedback divider **418** is a multi-modulus divider that in an embodiment can divide by N or N+1 according to the modulus control signal provided by the ΔΣ modulator. Of course, multi-modulus dividers with a wider range can also be used and are well known in the art.

[0025] By way of example to illustrate the source of quantization errors in fractional-N PLLs, FIG. **9** illustrates a first order ΔΣ modulator **900** that generates the modulus control signal for the feedback divider. The ΔΣ modulator **900** receives the fractional value m/n of the divide value at summer **902**. The integrator **904** accumulates the output of the summer and supplies quantizer **906**. When the accumulated value is greater than 1, quantizer **906** supplies a 1 and otherwise a 0. The quantizer output is the modulus control signal for the feedback divider and the quantizer output is also subtracted from the input m/n in summer **902**. Quantizer **906** supplies a stream of 1s and 0s with a 1 s density that equals the fractional value m/n as the modulus control signal. For example, a divide value being supplied to the divide control block **420** (see FIG. **4**) of 10 ¼ results in the divide control circuit supplying a divide control signal that averages 10 ¼ but has deterministic error. Thus, in the illustrated example the feedback divider divides by 10 or 11 with an average divide value of 10 ¼. For example, the modulus control signal causes the divider to divide by 10, 10, 11, and 10 to average 10 ¼. The deterministic error is the error between the actual divide value (N or N+1) and the average divide value of 10 ¼ resulting in a deterministic phase error being supplied to the loop filter by the TDC. That deterministic error is the quantization error resulting from the modulus control signal supplied by the quantizer **906**. Embodiments cancel the quantization error to improve the performance of the PLL. Note that while a first order ΔΣ modulator is illustrated here, other embodiments use a second or higher order ΔΣ modulator according to the needs of the system along with an appropriately matched multi-modulus divider.

[0026] FIG. **10** illustrates an embodiment of quantization error cancellation using the gain information from the TDC to more accurately cancel the quantization error. The gain information **405** supplied by TDC **402** (see FIG. **4**) is used to scale the quantization error **1002** in the scaling block **1004**. Thus, the TDC bits per DCO period is used to scale the quantization error. For example, if the 1-bit stream is 32 and the quantization error is 0.75, the quantization error is scaled to 24 and supplied as the scaled quantization error **1006** to the summer **1008**. When the number of delay elements in the **1**-bit stream changes due to voltage, temperature, or aging, the scaling changes. Summer **1008** subtracts the adjusted quantization error from the phase error (phase difference between fb_clk and ref_clk) supplied by the TDC. Since the gain information is continually calibrated against the period of the DCO output, the quantization error cancellation is also continually calibrated during regular operation of the PLL without having to take the PLL out of lock. Note that while the cancellation logic including scale logic **1004** and summer **1008** are shown as being separate from the loop filter **402** for ease of illustration, in embodiments they are incorporated into digital logic in the loop filter **410**.

[0027] While the gain information from the TDC is particularly useful to more accurately cancel quantization error in fractional-N PLLs, the gain information from the TDC can also be used to make the loop filter constants more accurately match the VCO gain in both integer N and fractional-N PLLs.

[0028] Thus, a PLL providing both gain and phase information from the TDC has been described. The description of the invention set forth herein is illustrative and is not intended to limit the scope of the invention as set forth in the following claims. The terms “first,” “second,” “third,” and so forth, as used in the claims, unless otherwise clear by context, is to distinguish between different items in the claims and do not otherwise indicate or imply any order in time, location, or quality. Variations and modifications of the embodiments disclosed herein may be made based on the description set forth herein, without departing from the scope of the invention as set forth in the following claims.

Claims

1. A method comprising: recovering phase information from an output of a time-to-digital converter (TDC), the phase information indicative of a phase error between a reference clock signal and a feedback clock signal; and recovering gain information from the output of the TDC, the gain information indicative of a relationship between a step size of the TDC and a period of a digitally controlled oscillator (DCO) output signal supplied by a digitally controlled oscillator (DCO).

2. The method as recited in claim 1 further comprising, generating the feedback clock signal as a pulse having a pulse width equal to a period of the DCO output signal.

3. The method as recited in claim 2 further comprising, recovering the phase information based on a transition between zeros and ones in the output of the TDC.

4. The method as recited in claim 3 further comprising, recovering the gain information based on length of the ones in the output of the TDC.

5. The method as recited in claim 1 further comprising adjusting a quantization error based on the gain information to generate an adjusted quantization error.

6. The method as recited in claim 5 supplying the adjusted quantization error to a loop filter for cancellation of the adjusted quantization error.

7. The method as recited in claim 1 further comprising using both a one to zero transition and a zero to one transition to determine the phase information.

8. A phase-locked loop (PLL) comprising: a time-to-digital converter (TDC) coupled to receive a reference clock signal and a feedback clock signal and configured to provide a TDC output indicative of a phase information between the reference clock signal and the feedback clock signal; a feedback divider coupled to a digitally controlled oscillator (DCO) output signal and configured to divide the DCO output signal and supply the feedback clock signal to the TDC; and wherein the TDC output further provides gain information, the gain information indicative of a relationship between a step size of the TDC and a period of the DCO output signal.

9. The PLL as recited in claim 8 wherein the feedback divider is configured to generate the feedback clock signal as a pulse with a pulse width of a period of a DCO output clock signal.

10. The PLL as recited in claim 9 wherein the TDC comprises a plurality of delay elements and the phase information is indicated by a transition between zeros and ones in the TDC output and the gain information is indicated as a number of Is in the TDC output, each of the Is corresponding to one of the delay elements.

11. The PLL as recited in claim 10 wherein the gain information is based on a number of delay elements in the TDC that correspond to the period of the DCO output clock signal.

12. The PLL as recited in claim 9 further comprising a delta sigma modulator and wherein a quantization error from the delta sigma modulator is adjusted by the gain information to provide adjusted quantization error.

13. The PLL as recited in claim 12 further comprising a summing circuit to subtract the adjusted quantization error from the phase information supplied by the TDC to thereby cancel the adjusted quantization error and generate adjusted phase information.

14. The PLL as recited in claim 13 further comprising a loop filter coupled to the TDC and configured to supply a control signal to the DCO based on the adjusted phase information.

15. An apparatus comprising: a digitally controlled oscillator (DCO) configured to supply a DCO output signal; a feedback divider coupled to the DCO output signal and configured to divide the DCO output signal and generate a feedback clock signal, the feedback clock signal having a pulse with a pulse width equal to a period of the DCO output signal; a time-to-digital converter (TDC) coupled to receive a reference clock signal and the feedback clock signal and configured to provide a TDC output indicative of a phase information between the reference clock signal and the feedback clock signal; and wherein the TDC output further provides gain information, the gain information indicative of a relationship between a step size of the TDC and a period of the DCO output signal.

16. The apparatus as recited in claim 15 wherein the TDC includes a plurality of delay elements and each of the delay elements corresponds to step size of the TDC.

17. The apparatus as recited in claim 16 wherein the gain information is indicated as a number of 1s in the TDC output, each of the Is corresponding to one of the delay elements.

18. The apparatus as recited in claim 15 further comprising a delta sigma modulator coupled to control the feedback divider and wherein a quantization error from the delta sigma modulator is adjusted by the gain information to provide an adjusted quantization error.

19. The apparatus as recited in claim 18 further comprising a summing circuit to subtract the adjusted quantization error from the phase information supplied by the TDC to thereby cancel the adjusted quantization error and generate adjusted phase information.

20. The apparatus as recited in claim 19 further comprising a loop filter coupled to the TDC and configured to supply a control signal to the DCO based on the adjusted phase information.